

SUMMARY

- Senior backend engineer, **5 years** building and owning **cloud-native distributed systems** in **Python and Go** on **GCP** and **AWS**.
- Currently owns both the **Go API service** and the **LLM decision engine** behind a multi-tenant retail planning platform — deciding which of **88,863** products a retailer keeps each season, over a **2.11B-row ClickHouse** fact table, with **agents held SELECT-only**.
- Ships and then hardens: **100% coverage CI gates**, **Kafka/Flink** ingestion at **30K+ menus/month**, and migrations that cut **p95 latency 4x**.

SKILLS

- Languages** – Python, Go (Golang), SQL.
- Backend & APIs** – FastAPI, Gin, Django, REST, Pydantic, asyncio, goroutines, Celery, Google Wire (DI).
- AI & Agents** – LLM agents (OpenAI/Anthropic/Gemini), LangGraph, LangChain, tool calling, prompt engineering, RAG, embeddings, Milvus, offline evaluation, model benchmarking, LangSmith.
- Distributed Systems** – Data structures & algorithms, system design (HLD/LLD), microservices, concurrency, bounded parallelism, idempotency, dead-letter queues, caching, sharding, reliability/SLOs.
- Streaming & Data** – Apache Kafka, Apache Flink, ClickHouse, BigQuery, ETL, batch processing.
- Databases & Storage** – PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, S3, GCS.
- Cloud, CI/CD & Security** – GCP, AWS, Docker, Kubernetes, Bazel, Bitbucket Pipelines, golangci-lint, SAST/SBOM, JWT/OIDC, RBAC, Secret Manager, Datadog, New Relic, ELK, Linux.

PROFESSIONAL EXPERIENCE

Senior Software Engineer IMPACT ANALYTICS, Bangalore May 2026 – Present

AssortSmart — AI planning platform that decides which products retailers stock each season

- Own the Go service every tenant hits for planning APIs** — **36 REST routes** across **8 modules** behind one **Google Wire** compile-time injection root. It runs as the HTTP edge with **no reverse proxy in front**, so it enforces its own protection rather than inheriting it: HTTP/2 cleartext (**h2c**), nested request timeouts, a 10 MiB body cap, and graceful **SIGTERM** draining.
- Keep regressions out of that service** with a **100% statement-coverage gate** over **1,200+ Go tests** that fails a local push and the CI pipeline alike, plus **golangci-lint**, race detection, and **SAST/SBOM** scanning on merges; author and maintain its onboarding, architecture, and auth **documentation**.
- Enforce tenant isolation on every request** so no retailer can read another's stores or products — **Firestore Admin** or local **JWT** verification falling back to **Google OIDC** public keys, tenant resolved from header or hostname, and constant-time internal API keys.
- Built the engine that decides which products a retailer keeps or drops next season** — **88,863 products** scored per full-season pass at under **\$1K** in model spend — blending a deterministic sell-through and rate-of-sale score with **8 weighted LLM checks** — colour cannibalisation, size-curve health, trend velocity, margin resilience among them — run in bounded parallel.
- Ensured a hallucinated query can never write to the planning warehouse**, holding agents **SELECT-only** on a **2.11B-row ClickHouse** table through **five independent controls** — prompt ban, a typed no-raw-SQL tool contract, a write-statement guard, per-query budgets, and a config assertion that **fails the run if writes are enabled** — with all mutations staged as **CSV** and promoted by hand.
- Made every merchandising decision auditable without re-running the pipeline**, shipping a read-only planner assistant that explains why any style was kept or dropped — a **LangGraph** tool-calling loop bounded to 4 read-only calls per question over frozen, versioned decision snapshots.
- Took the planning grid from unusable to interactive** — **189s to 12.3s (~15.5x)** on **250M rows** — by benchmarking **ClickHouse** against **PostgreSQL** on row-identical data, then leading its adoption as the platform's analytics store.
- Tech** – Go, Gin, Google Wire, Python, FastAPI, LangGraph, ClickHouse, BigQuery, Redis, GCS, GCP, Docker, Datadog, LangSmith.

Software Development Engineer 2 UBER (via EPAM Systems), Bangalore July 2024 – May 2026

Financial Risk Management — backend deciding which of Uber's accounts get audited each quarter

- Owned the risk-scoping backend** behind Uber's quarterly financial close — **30+ REST APIs** across **8** screens in Python, FastAPI, and MySQL — targeting a **70%** reduction in manual reconciliation, from about **2 weeks** to **3–4 days**.

- Encoded Uber’s in-scope scoping rules** across ~**55** statement line items and **14** entities in the quarterly close, implementing materiality, qualitative-override, residual-risk, and **5%** component-threshold logic against **\$340M** materiality and a **\$170M** residual threshold.
- Migrated quarterly reconciliation from Google Sheets to MySQL** in an **18-file** change across five layers, shipping parallel /**v2** APIs that build the three-level statement-line-item hierarchy, reconcile internal **HFM** balances against the published **10-Q**, and assign every line a stable ID for audit.
- Led 3 engineers** on the backend, owning API contracts, **code review**, and layered **design reviews** across the scoping services.
- Tech** – Python, FastAPI, SQLAlchemy 2.0, Pydantic, MySQL, Bazel, Docker.

Menu Ingestion & Automation — getting restaurant menus onto Uber Eats without humans typing them

- Cut restaurant onboarding from 24 hours to 2 hours** at **30K+ menus/month**, owning ingestion end to end on **GCP**: **Selenium** acquisition, a **Kafka** ingest bus, and **Flink** keyed normalize/dedupe.
- Converted multilingual PDF and photo menus into catalog records at 98% field fidelity** and **100% schema validity** (offline eval), building a **LangChain RAG** path over **Milvus** embeddings with **Gemini 2.5 Pro** and a blocking schema-validation gate before catalog upsert.
- Raised ingestion success to 95%+** by hardening the scrape fleet with IP rotation, dynamic proxy pools, and adaptive retries.
- Tech** – Python, Selenium, Kafka, Flink, LangChain, Gemini, RAG, Milvus, GCP, Docker.

ANZ Driver Document Compliance (Uber Mobility)

- Reached 99.9% document compliance** and eliminated ~**20 hours/week** of manual verification by automating driver and vehicle document checks for **ANZ** against local authority requirements.

Software Development Engineer 2

MASTERS INDIA, Noida

December 2022 – June 2024

GST Compliance & E-Invoicing — filing tax invoices with the Indian government at scale

- Cut p95 latency from 1.2s to 300ms** for **1,500+ clients** by migrating a **PHP/Laravel** monolith to **FastAPI** microservices, while mentoring **2 engineers**.
- Raised throughput from 700 to 4,000 requests/min** — sustaining **1M+ government submissions/day** and **100K+** per bulk import — on a **Kafka** and **PostgreSQL** pipeline with tables sharded by tax quarter.
- Made government-portal filings safe to retry** with **idempotency keys**, bounded retries, and a **dead-letter queue**, and cut repeat DB reads **30%** with **Redis** caching.
- Cut incident triage and debugging time 70%, support tickets 35%** with **ELK** and **New Relic** alerting plus a client usage dashboard, raising test coverage from **35% to 82%** at **98%** deploy success.
- Tech** – Python, FastAPI, Kafka, PostgreSQL, MongoDB, Redis, Elasticsearch, Celery, Docker, ELK, New Relic, AWS.

Software Development Engineer

GEEKSFORGEEKS, Noida

August 2021 – November 2022

- Stabilized doubt-support under 10K+ daily queries** and **10x contest spikes** by migrating the backend from **PHP** to **Django**, and saved ~**70%** of the manual effort on those ops workflows with scheduled jobs for video processing, reminders, and cleanup.
- Lifted premium subscriptions 15–20%** by shipping voting, pinning, and locking REST APIs on **MySQL**, **MongoDB**, **Redis**, and **Elasticsearch**, and **raised course sales 30%** with an influencer earnings and analytics dashboard.

EDUCATION

B.Tech (Information Technology)

Institute of Engineering & Technology, Lucknow

July 2017 – June 2021

ACHIEVEMENTS & CERTIFICATIONS

- Google Code Jam 2021** — ranked **2260 / 37,000+** in the qualifier. **Smart India Hackathon 2020 (MHRD)** — national finalist, top 3. **LangChain Academy** certified.